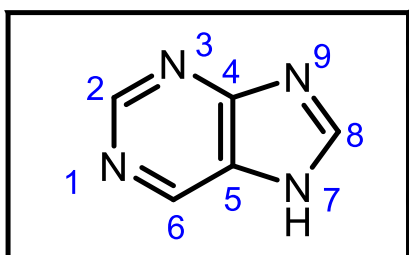
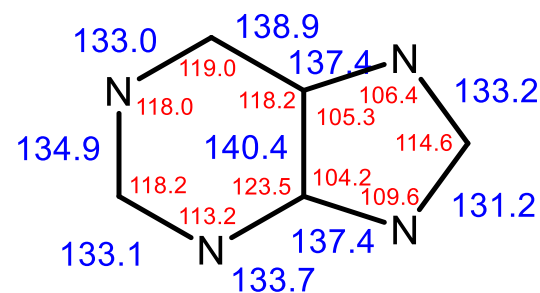


5.4. Anellierte Sechsring-Heterocyclen mit mehreren Heteroatomen

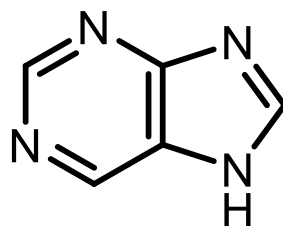
5.4.1. Purine



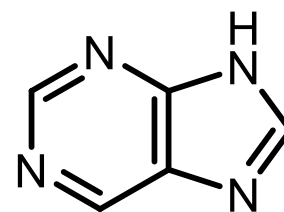
Bindungslängen [pm]
Bindungswinkel [°]



Tautomerie



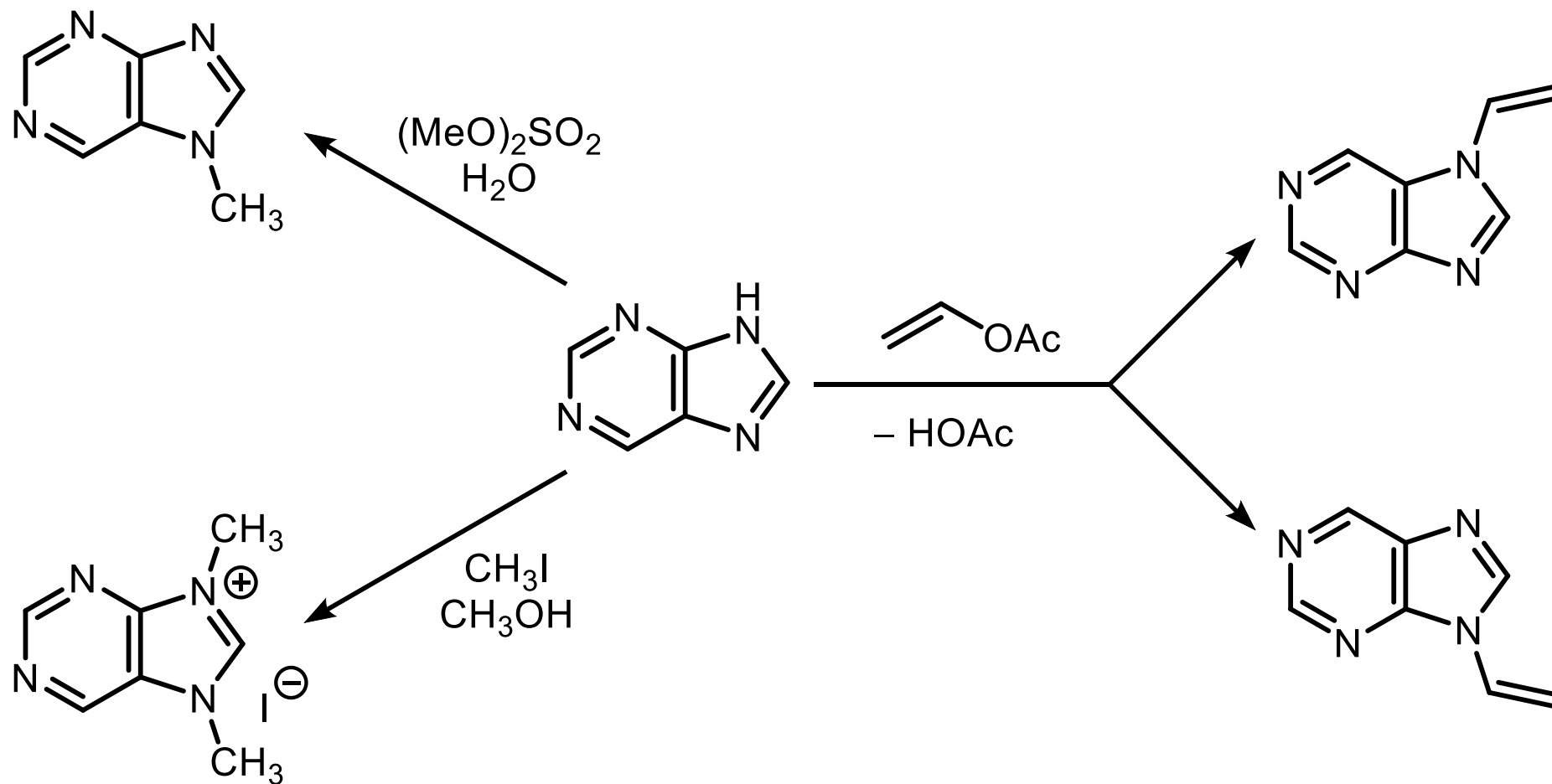
7*H*-Purin



9*H*-Purin

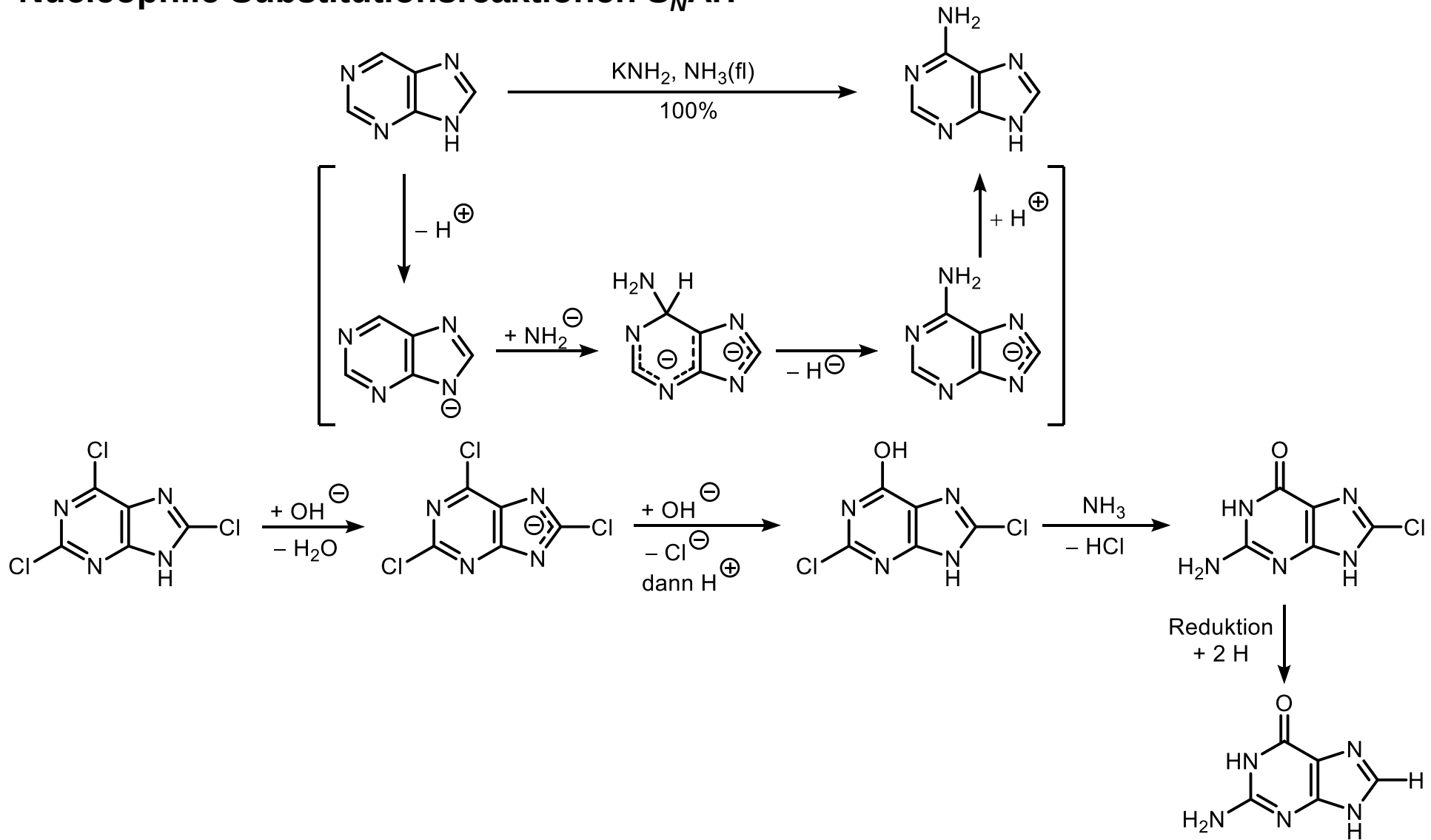
5.4.1. Purine Reaktionen

Mit Elektrophilen: an den N-Atomen

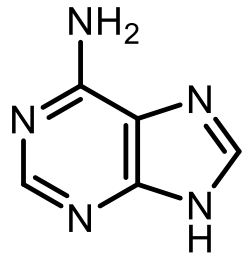


5.4.1. Purine Reaktionen

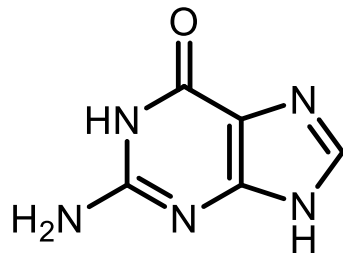
Nucleophile Substitutionsreaktionen S_NAr :



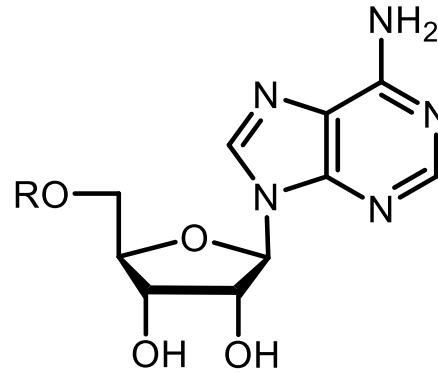
5.4.1. Purine Vorkommen



Adenin
("Purinbasen" der Nucleinsäuren)



Guanin

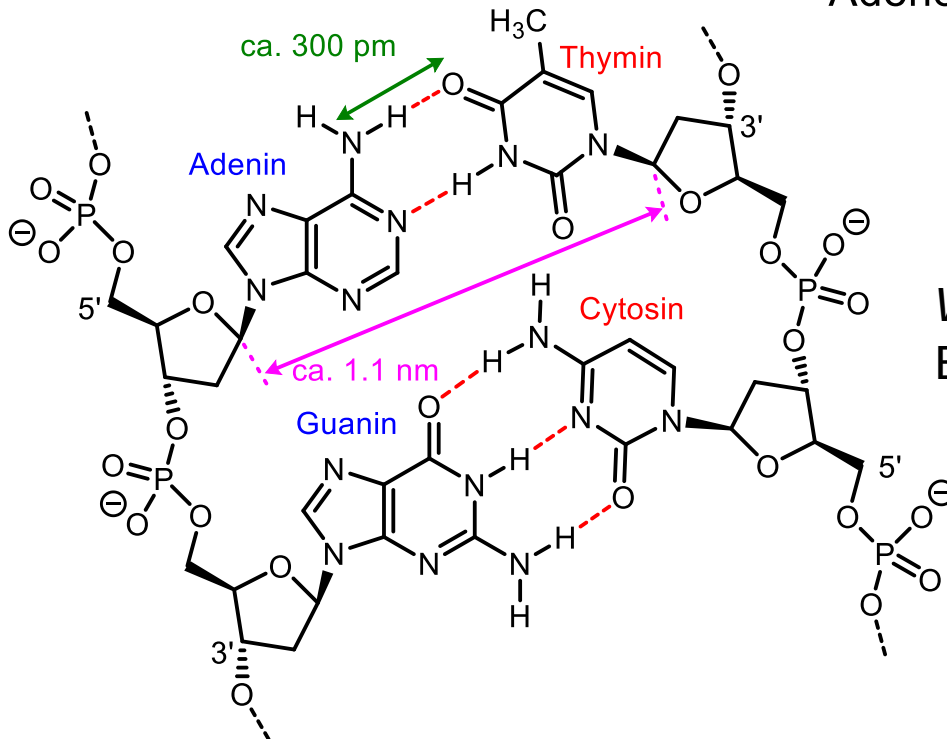


Adenosin (R = H)

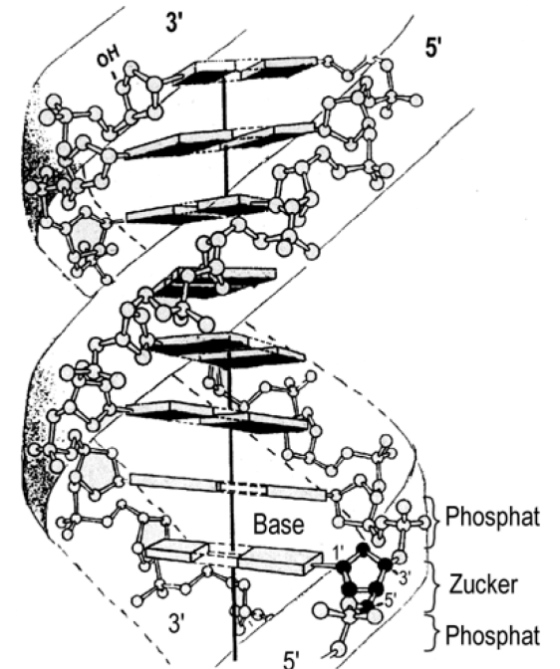
Adenosinmonophosphat (AMP)
(R = P(O)(OH)₂)

Adenosindiphosphat (ADP)
(R = P(O)(OH)OP(O)(OH)₂)

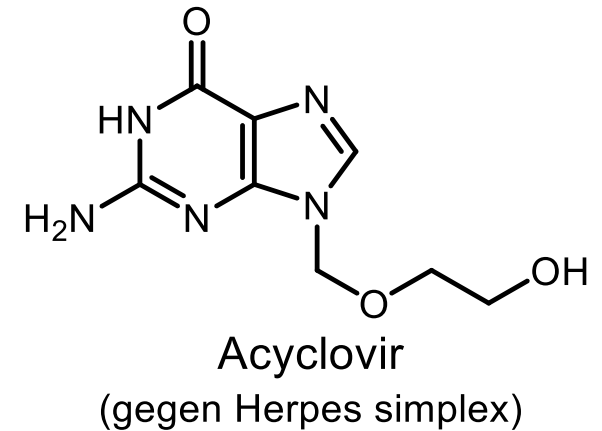
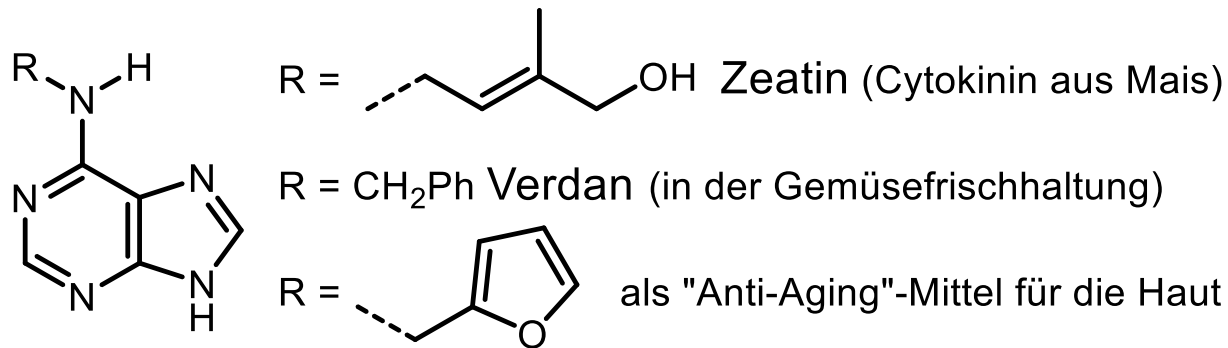
Adenosintriphosphat (ATP)
(R = P(O)(OH)OP(O)(OH)OP(O)(OH)₂)



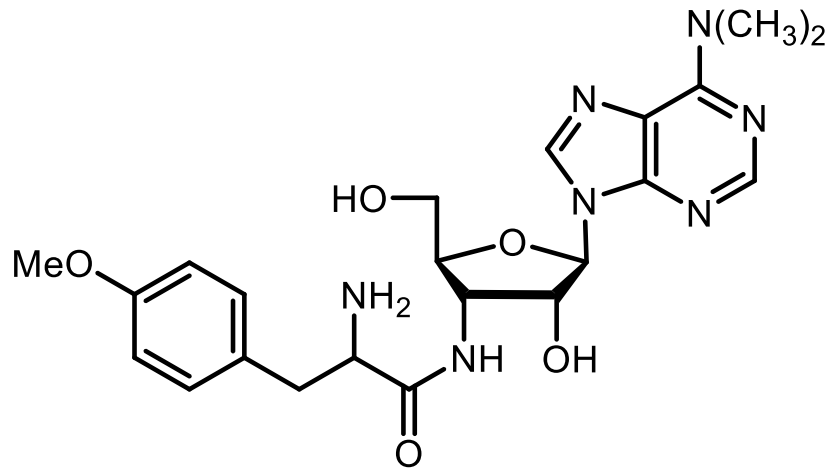
Watson-Crick-
Basenpaarung



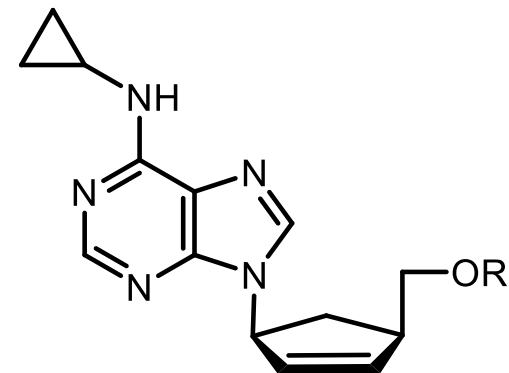
5.4.1. Purine Wirkstoffe



Nucleosid-Antibiotika

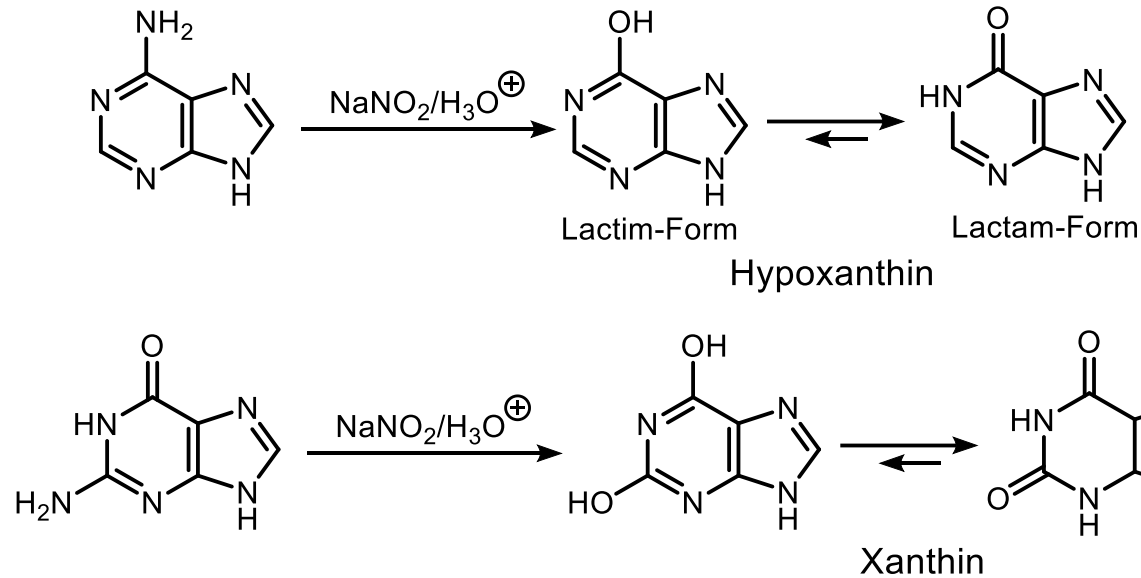


Puromycin
(Inhibitor der Proteinbiogenese in
Bakterien und Säugetierzellen)

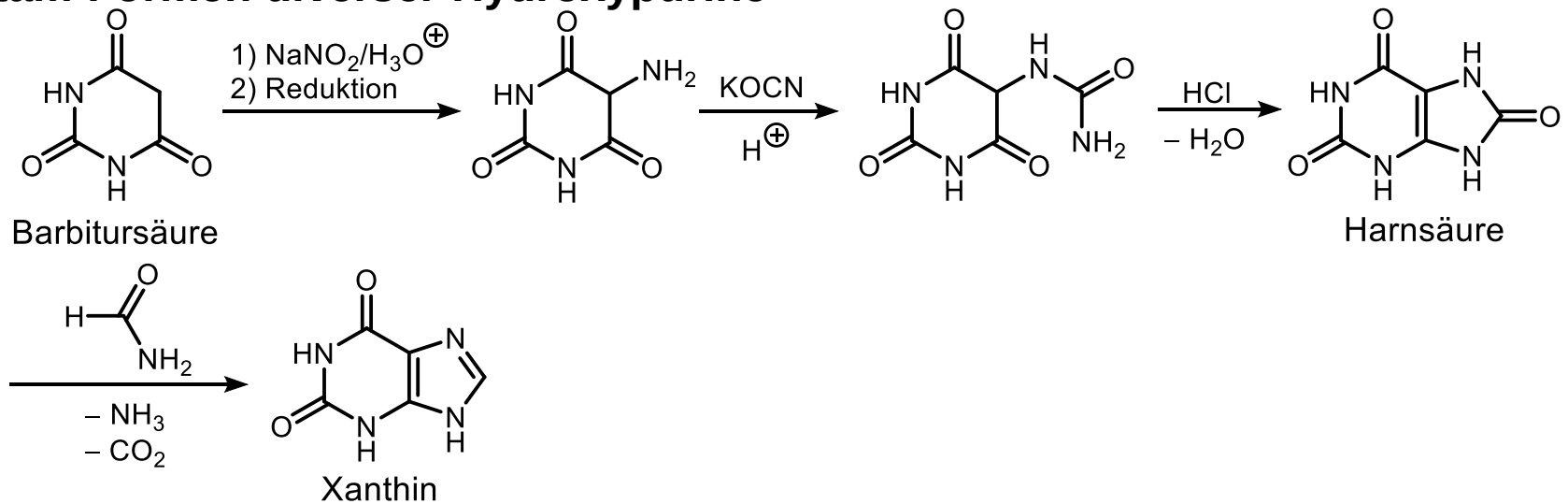


Abacavir
(potentes anti-HIV-Mittel)

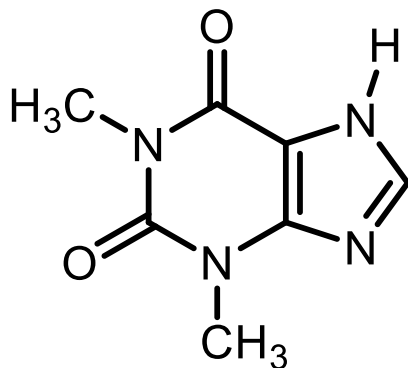
5.4.1. Purine Hydroxypurine



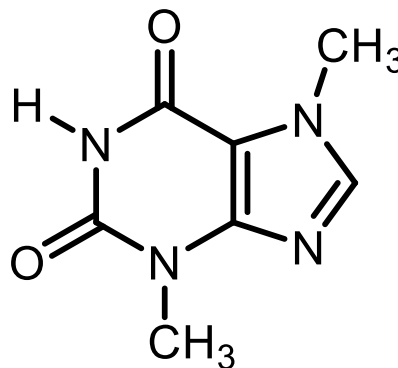
Lactam-Formen diverser Hydroxypurine



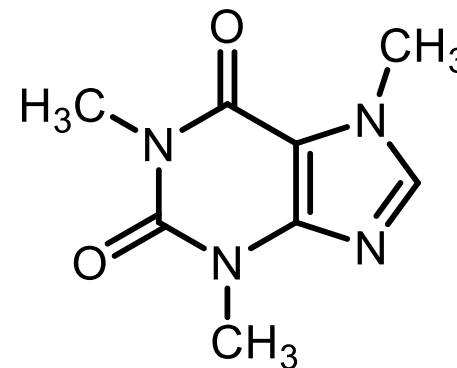
5.4.1. Purine Purinalkaloide



Theophyllin



Theobromin



Coffein

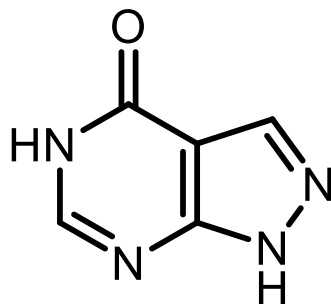
Coffein (= 1,3,7-Trimethylxanthin)

Isolierung: Runge, 1820.

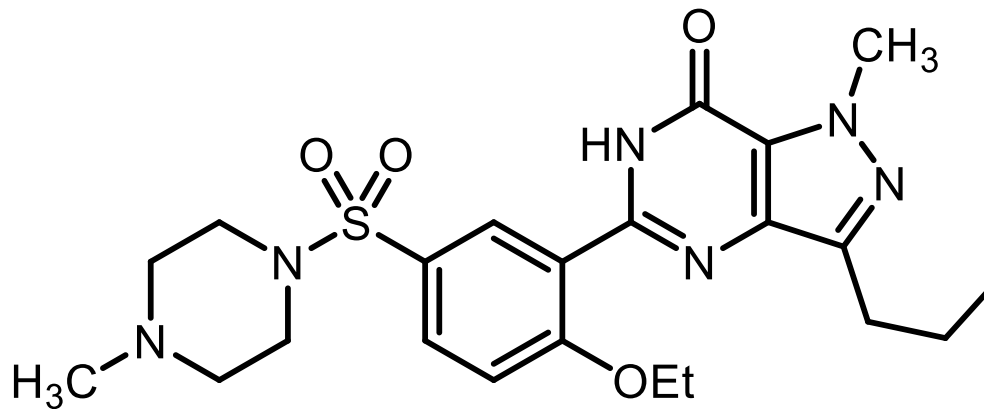
Coffein hemmt die Phosphodiesterase, die cyclisches AMP (Adenosinmonophosphat) in AMP umwandelt, besitzt somit eine erregende Wirkung auf das ZNS. Herztätigkeit, Stoffwechsel und Atmung werden angeregt. Blutdruck und Körpertemperatur steigen, die Blutgefäße im Gehirn erweitern sich, die der Eingeweide verengen sich, weshalb von Müdigkeit verschont wird, sich die Stimmung hebt und die Leistungsfähigkeit gesteigert wird. Coffein ist auch bekannt für seine diuretische (harntreibende) Wirkung.

5.4.1. Purine Isostere Wirkstoffe

Isostere des Purins: z. B. Pyrazolopyrimidine



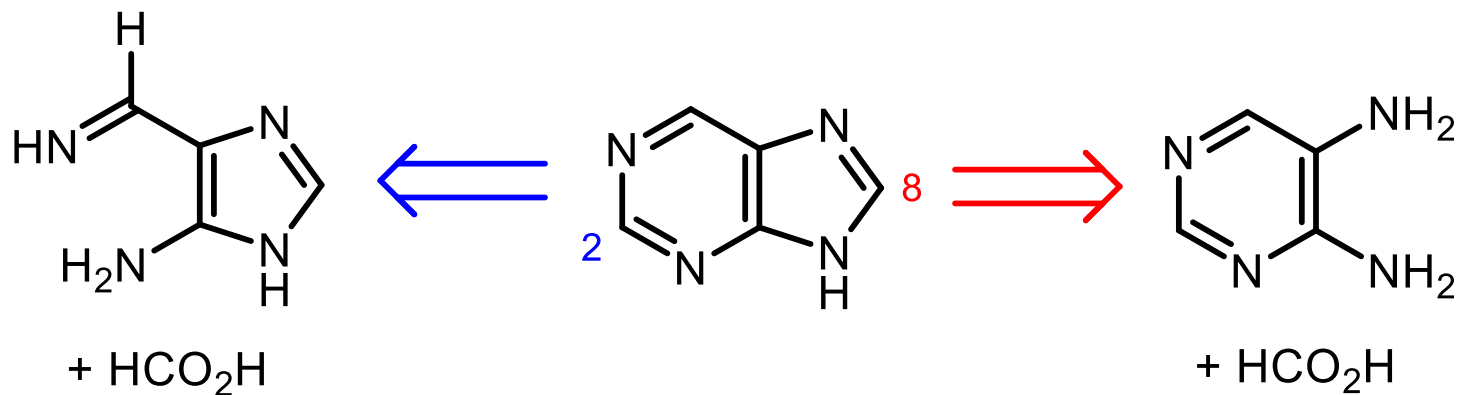
Allopurinol
(Behandlung von Gicht)



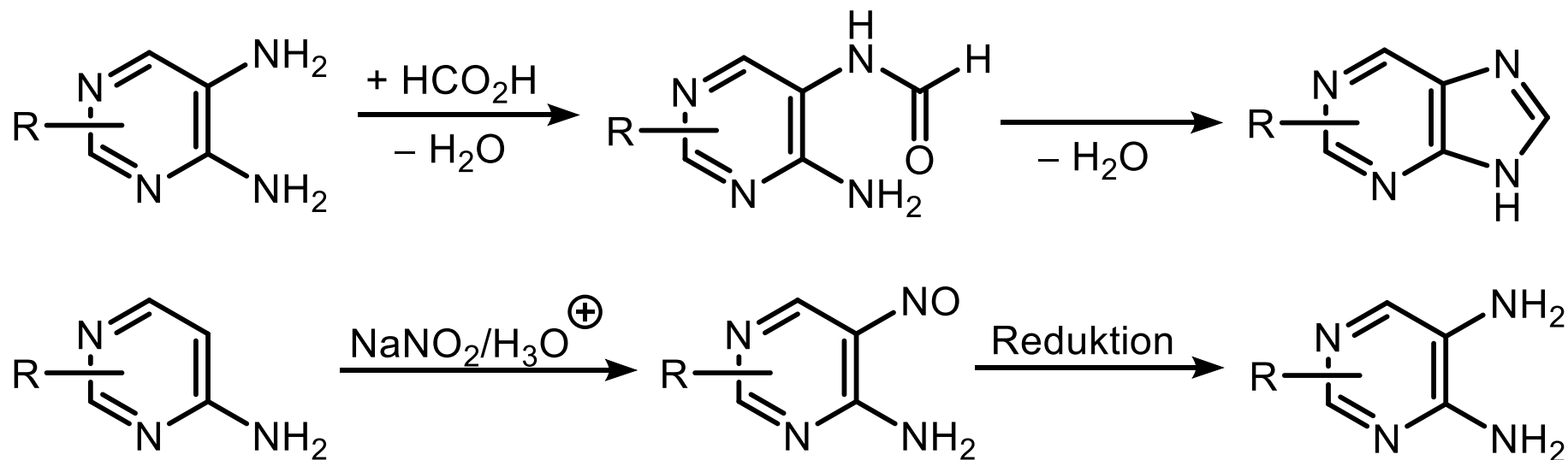
Sildenafil
(Inhibitor der Typ V cGMP-Phosphodiesterase)

5.4.1. Purine Synthesen

Retrosynthese

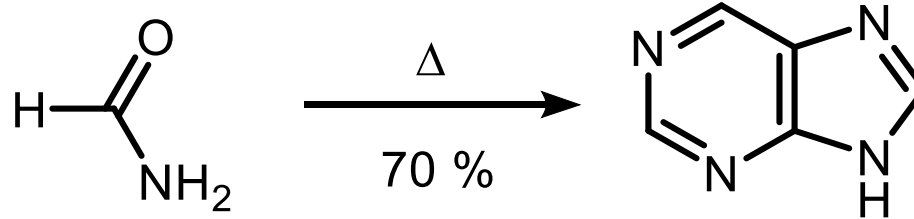


1) Traube-Synthese (1910)

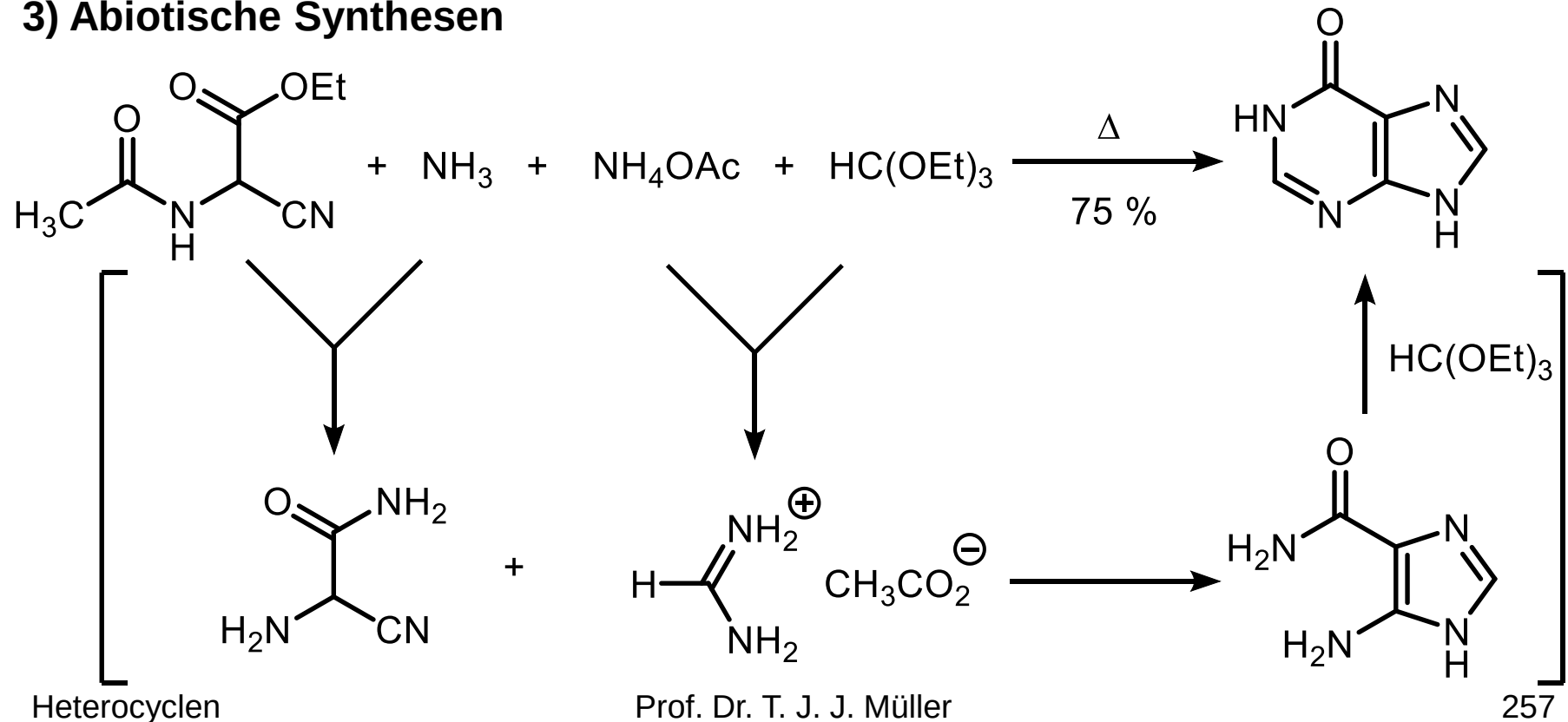


5.4.1. Purine Synthesen

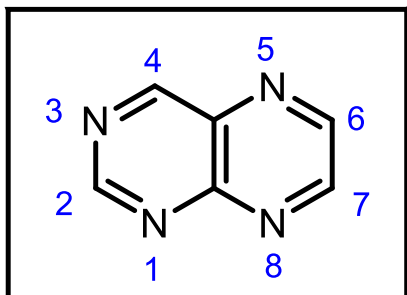
2) aus einfachen acyclischen Bausteinen



3) Abiotische Synthesen

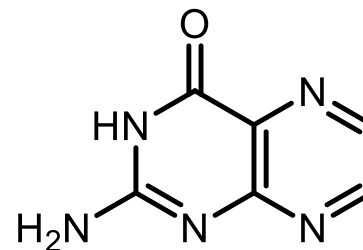


5.4.2. Pteridine

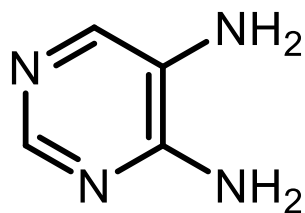
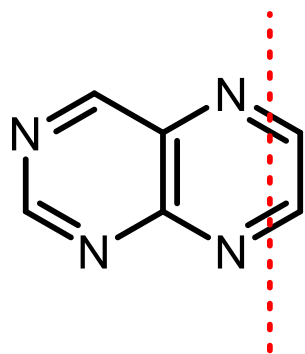


Pteridin (gelb)

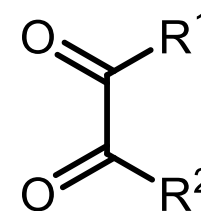
Formal: Pyrimidin + Pyrazin (IUPAC: Pyrazino[2,3-*d*]pyrimidin)
Extrem elektronenarm (Tieffeldverschiebung H, C im NMR)
Vorkommen: Farbstoffe der Schmetterlinge (Pterine), Coenzyme



Pterin



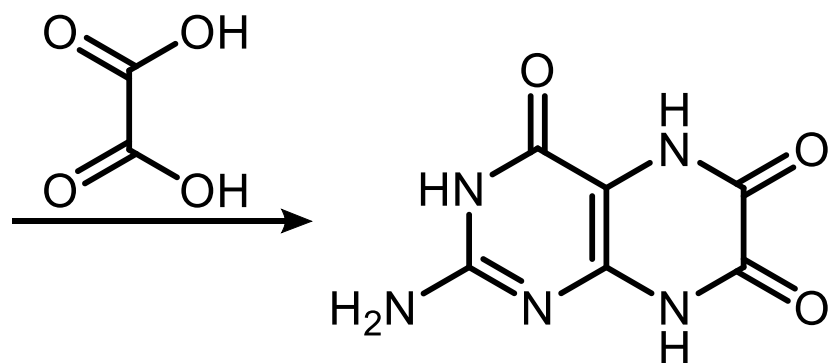
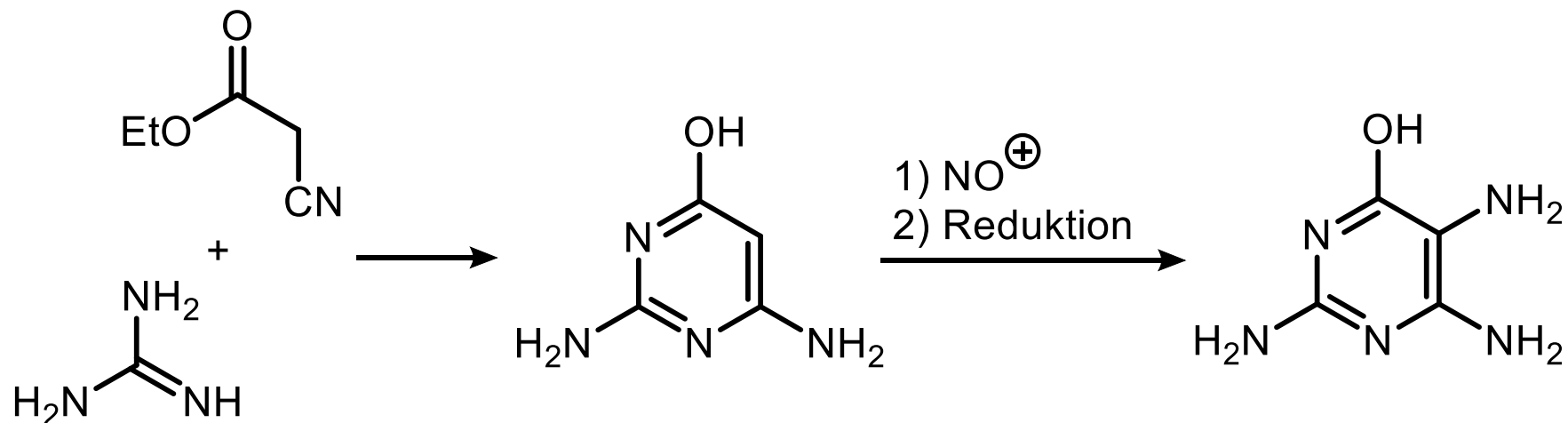
4,5-Diaminopyrimidin



1,2-Dicarbonylverbindung

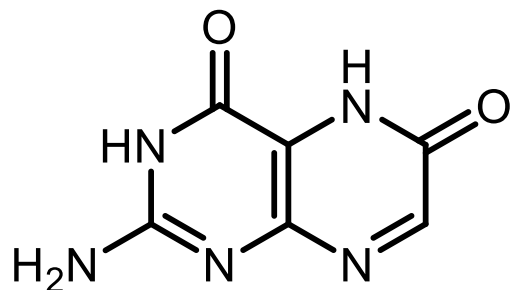
5.4.2. Pteridine Synthesen

Leukopterin (weißer Farbstoff des Kohlweißlings, 1926 aus 200.000 (!) Schmetterlingen isoliert)

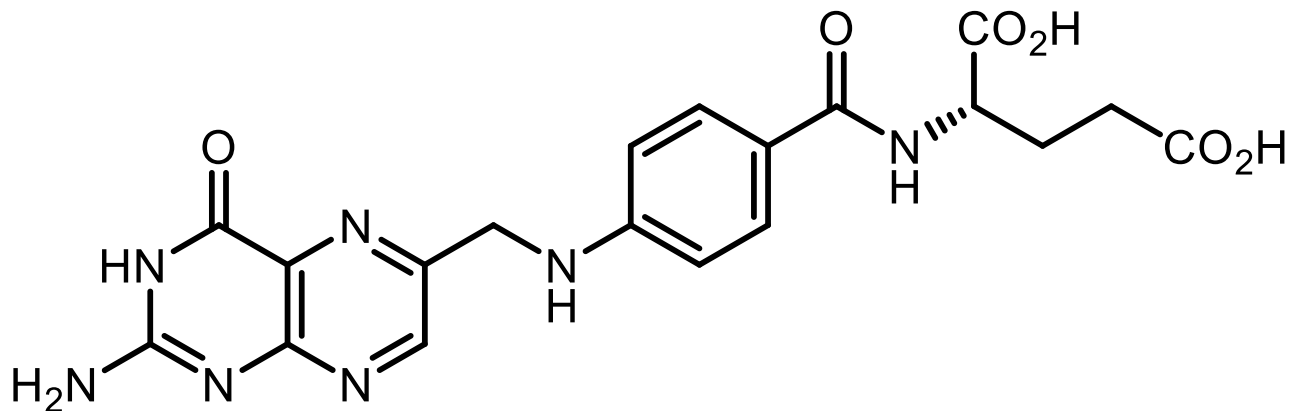


Leukopterin
(weiß)

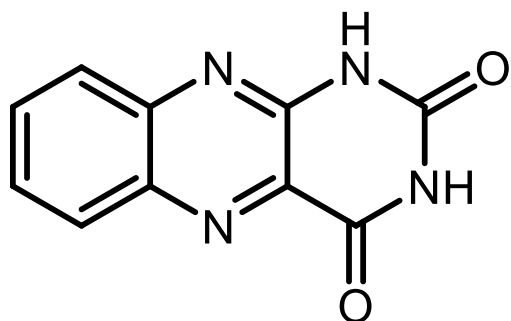
5.4.2. Pteridine Vorkommen



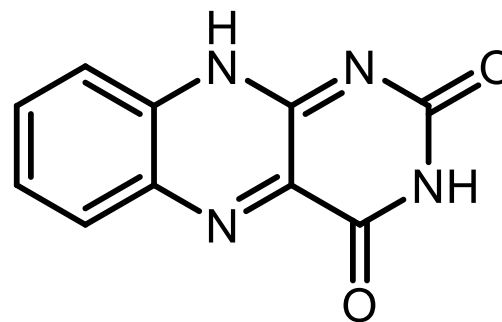
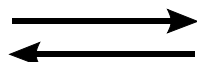
Xanthopterin
(gelb, Zitronenfalter)



Folsäure
(orange, Vitamin der B-Gruppe)



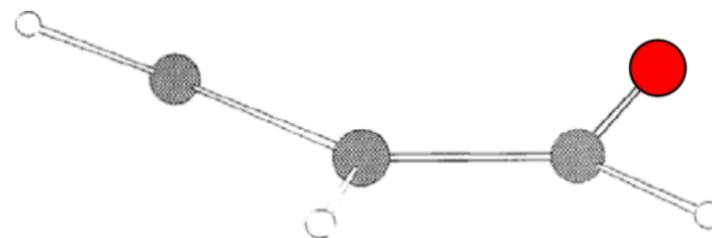
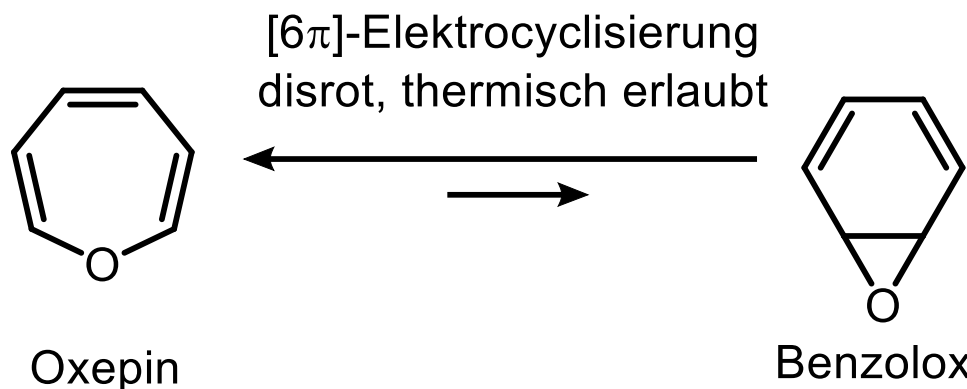
Alloxazin



Isalloxazin (Flavin)

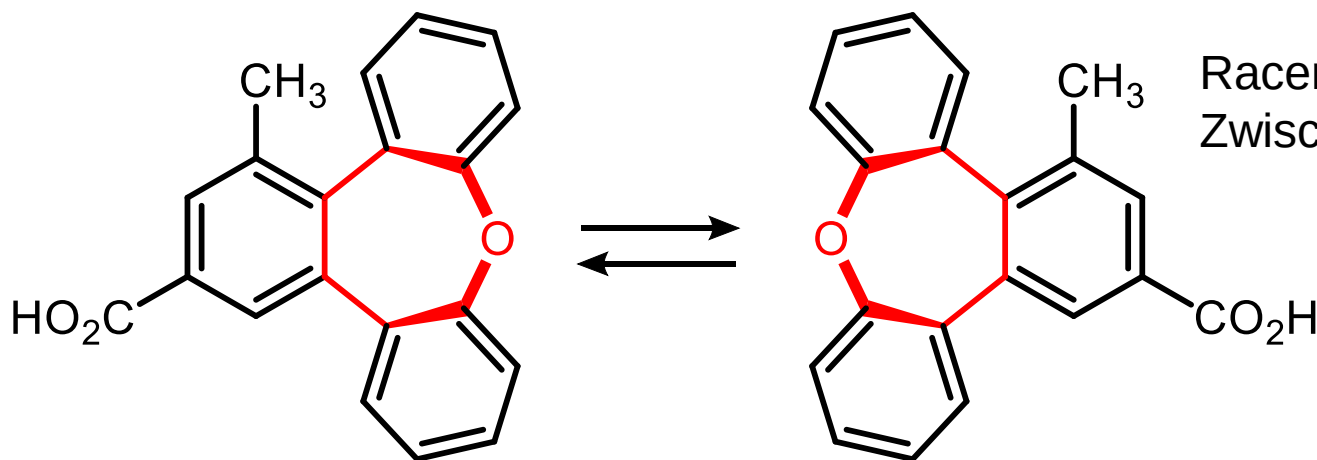
6. Ausgewählte Siebenring-Heterocyclen

6.1. Oxepin



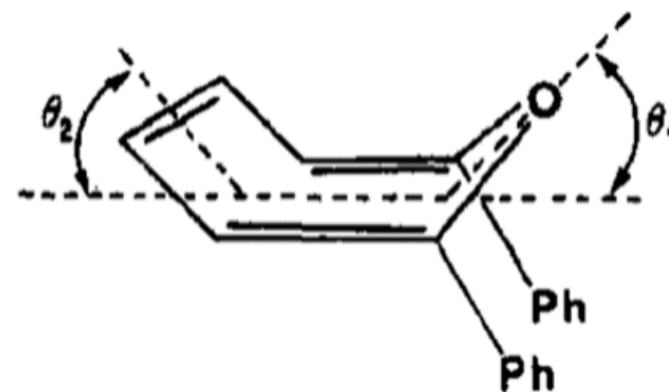
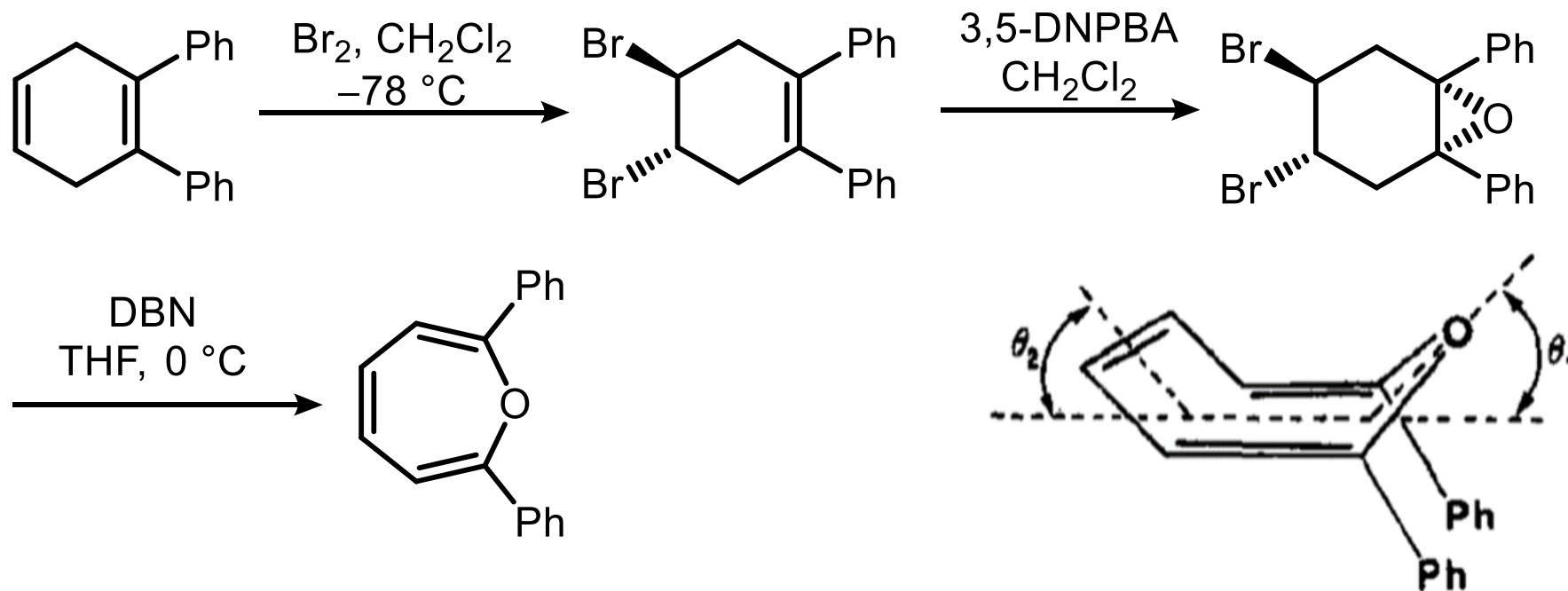
Toyota et al.
J. Phys. Chem. A **2003**, 107, 2749.

8 π , lokalisierte C=C-Bindungen (Bindungslängenalternanz), gewinkelt (Boot-C₅-Geometrie) $\Rightarrow$ Nicht-Aromat

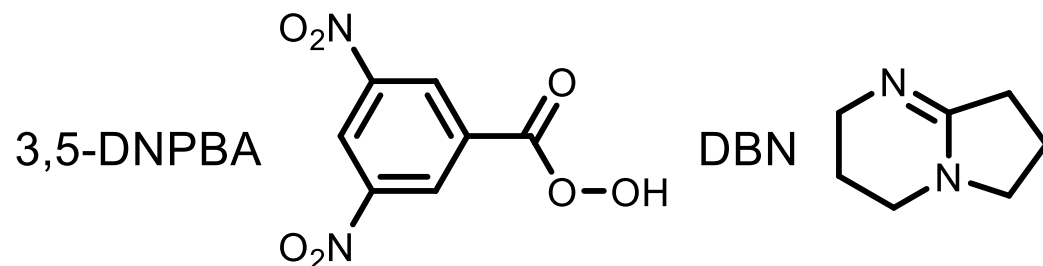


6.1. Oxepin Synthese

Synthese von 2,7-Diphenyloxepin

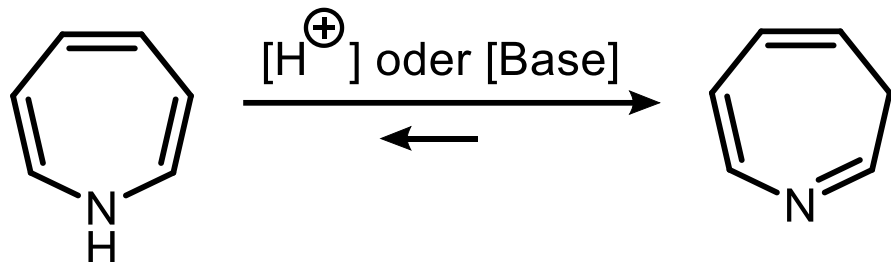


Röntgenstrukturanalyse



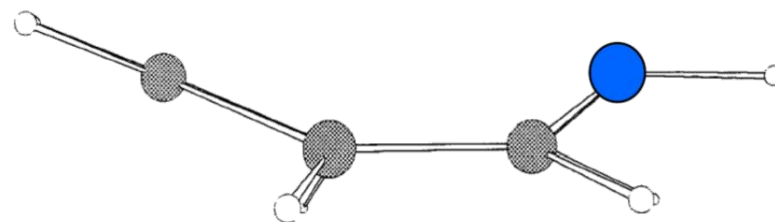
Boyd et al., *J. Org. Chem.* **1986**, 2784.

6.2. Azepin



1*H*-Azepin
(instabil)

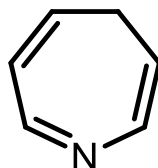
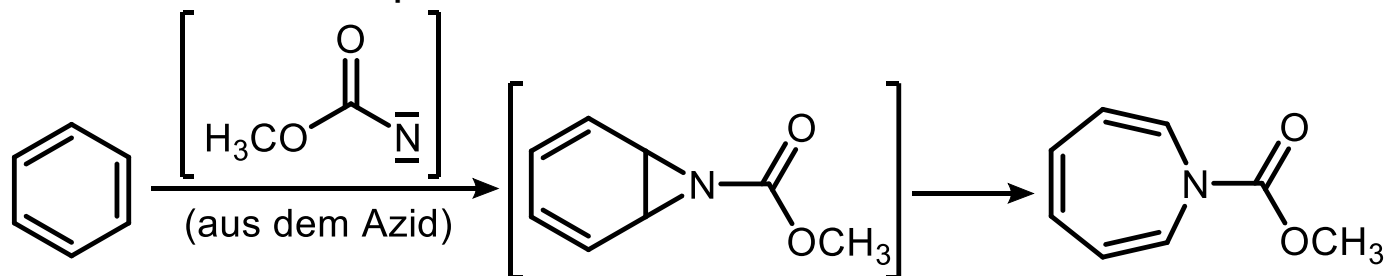
3*H*-Azepin



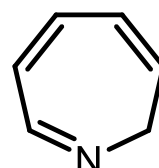
Toyota et al.
J. Phys. Chem. A **2003**, 107, 2749.

8 π , lokalisierte C=C-Bindungen (Bindungslängenalternanz), gewinkelt (Boot- C_s -Geometrie) $\Rightarrow$ Nicht-Aromat

N-Amid-substituierte 1*H*-Azepine sind stabil



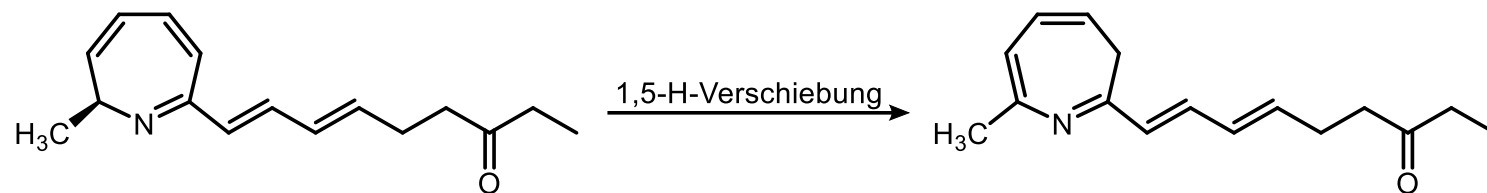
4*H*-Azepin



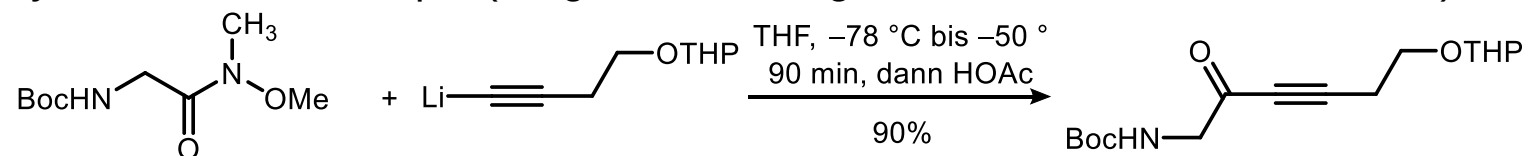
2*H*-Azepin

6.2. Azepin

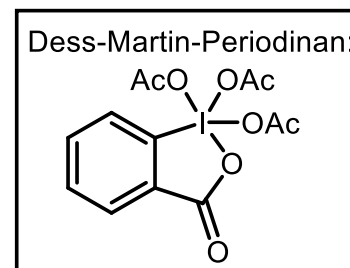
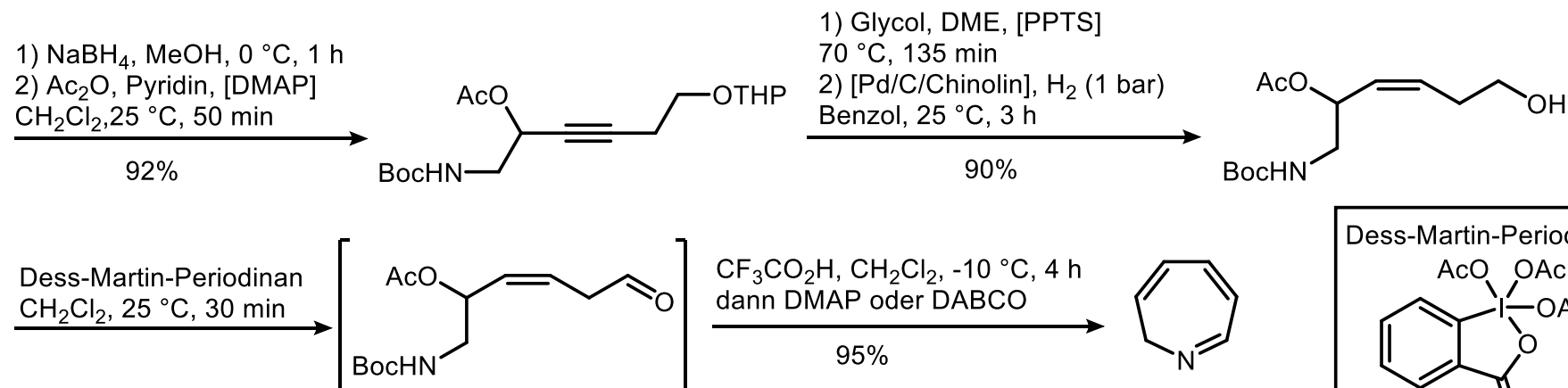
2H-Azepin als Teilstruktur des Naturstoffs Chalciporon aus dem Pfefferröhrling (*Chalciporus piperatus*)



Synthese von 2H-Azepin (Steglich et al., *Angew. Chem. Int. Ed.* **1995**, 1469.)

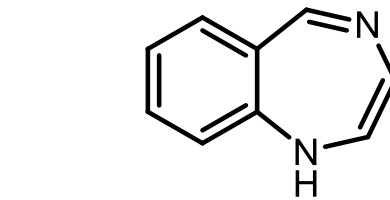
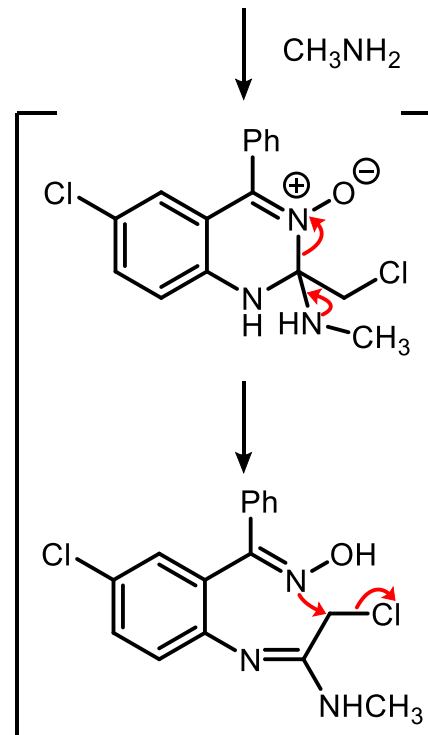
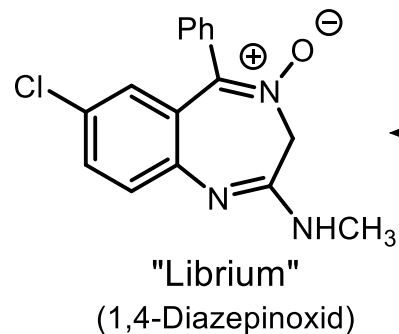
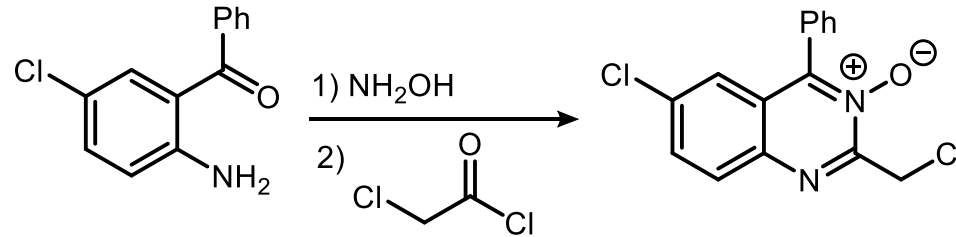


Weinreb-Amid des Glycins



6.3. Diazepin

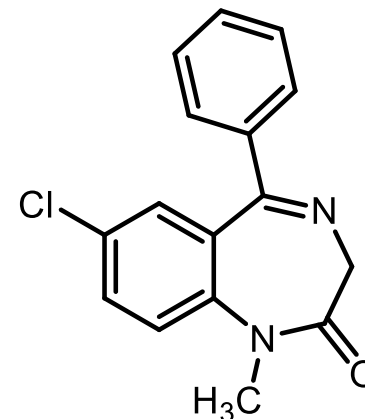
Leo Sternbach, Roche (1950er)



Benzo-1,4-diazepin

Tranquilizer

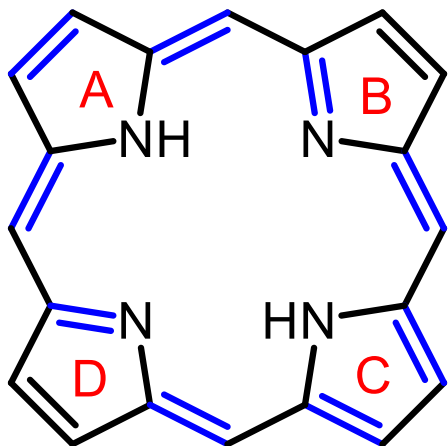
Benzodiazepine wirken im Allgemeinen: anxiolytisch (angstlösend), antikonvulsiv (krampflösend), muskelrelaxierend (muskelentspannend), sedativ (beruhigend und schlaffördernd), amnestisch (Erinnerung für die Zeit der Wirkdauer fehlt) und leicht stimmungsaufhellend.



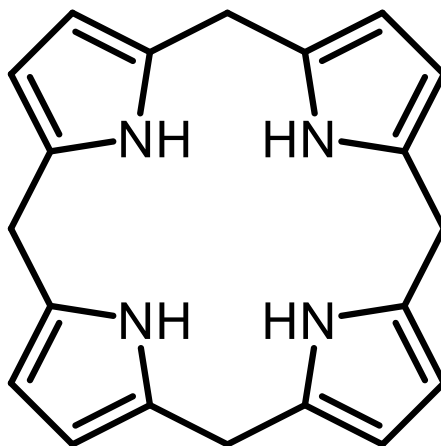
Diazepam
(Valium, 10 x aktiver)

Benzodiazepine gelten als die Medikamente mit der höchsten Missbrauchsrate in Deutschland.

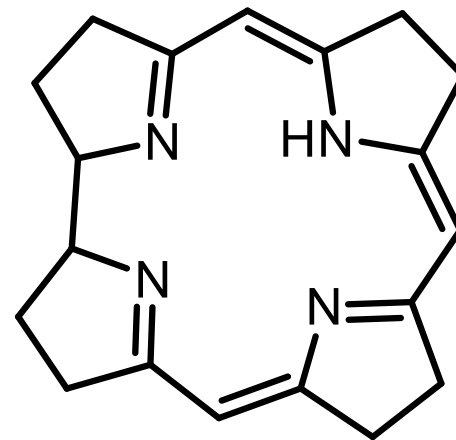
7. Porphyrin – ein makrocyclischer Heterocyclus



Porphyrin
(18π -System)

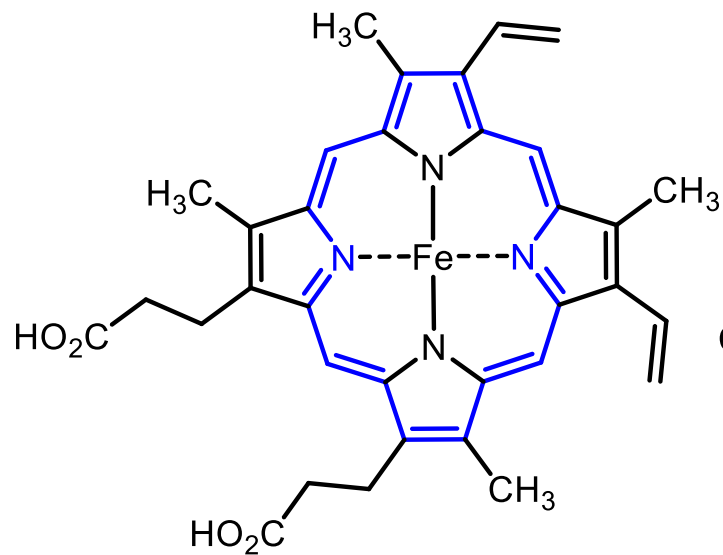
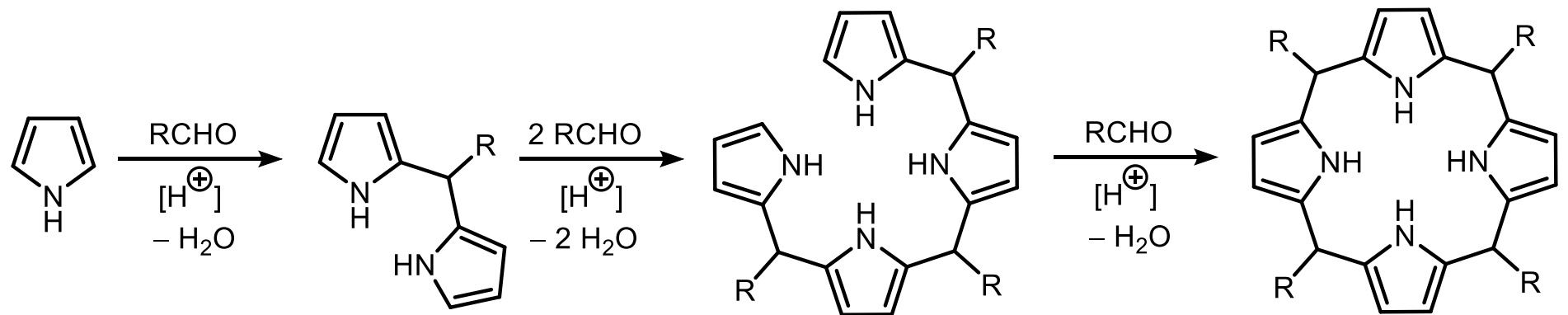


Porphyrinogen
(Hexahydroporphyrin)

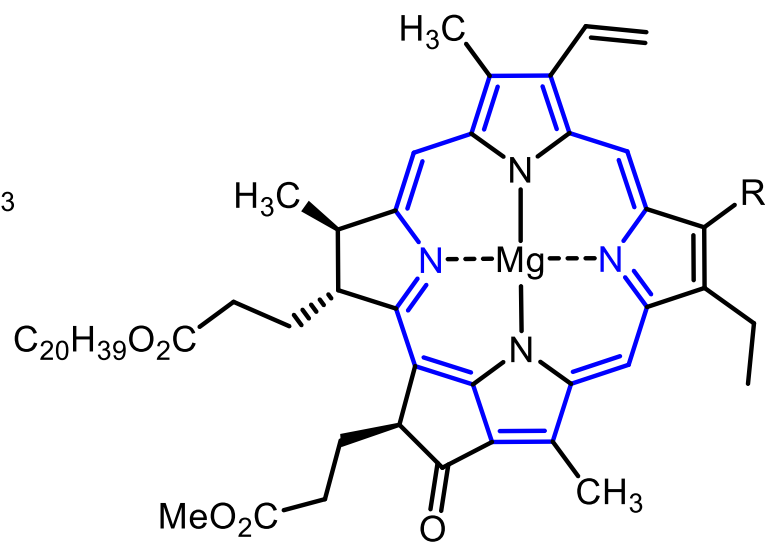


Corrin

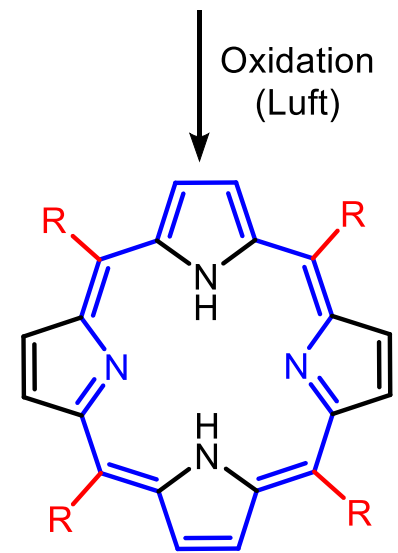
7. Porphyrin



Häm

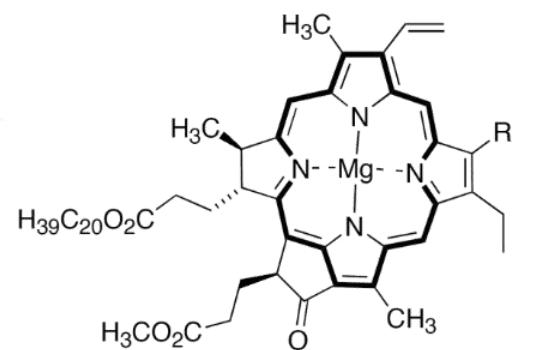
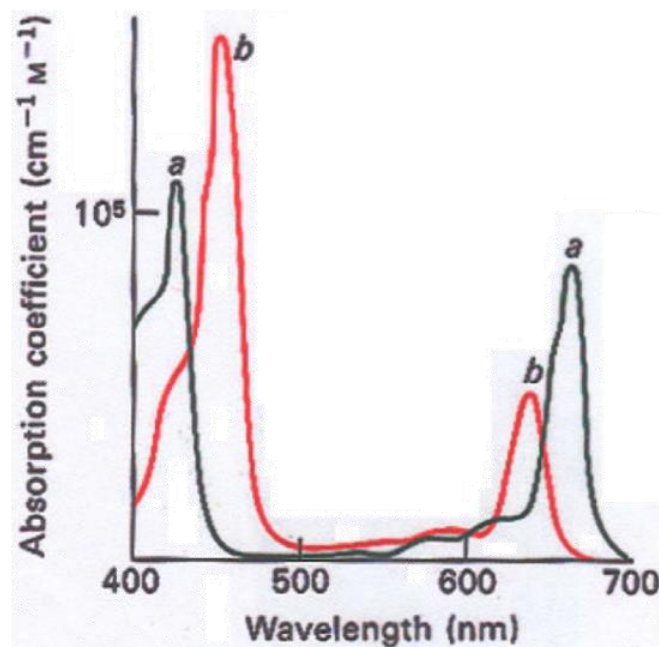
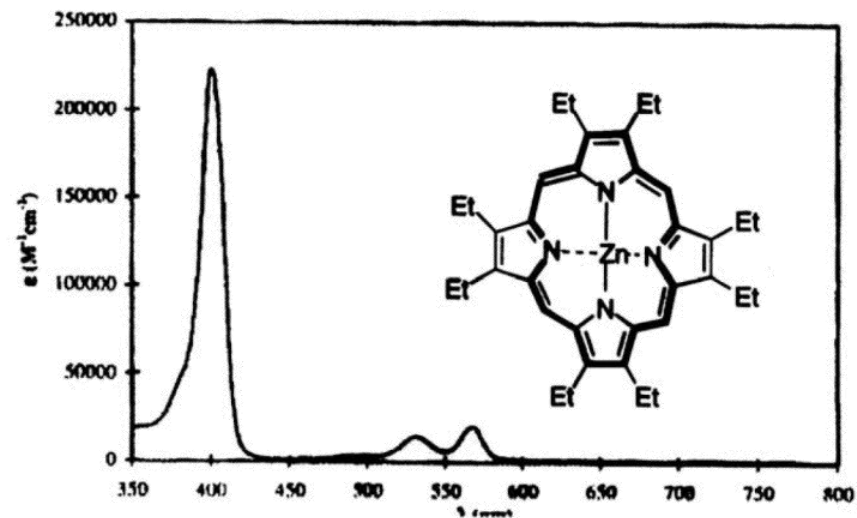
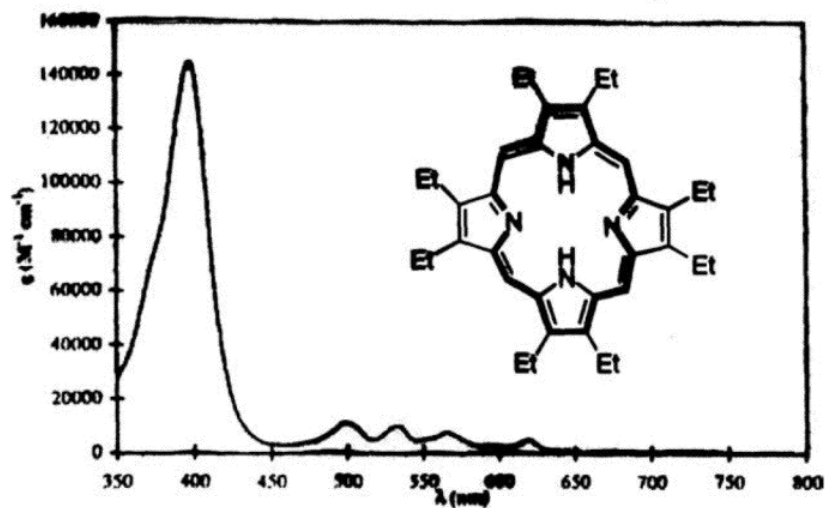


Chlorophyll a (R = CH₃)
Chlorophyll b (R = CHO)



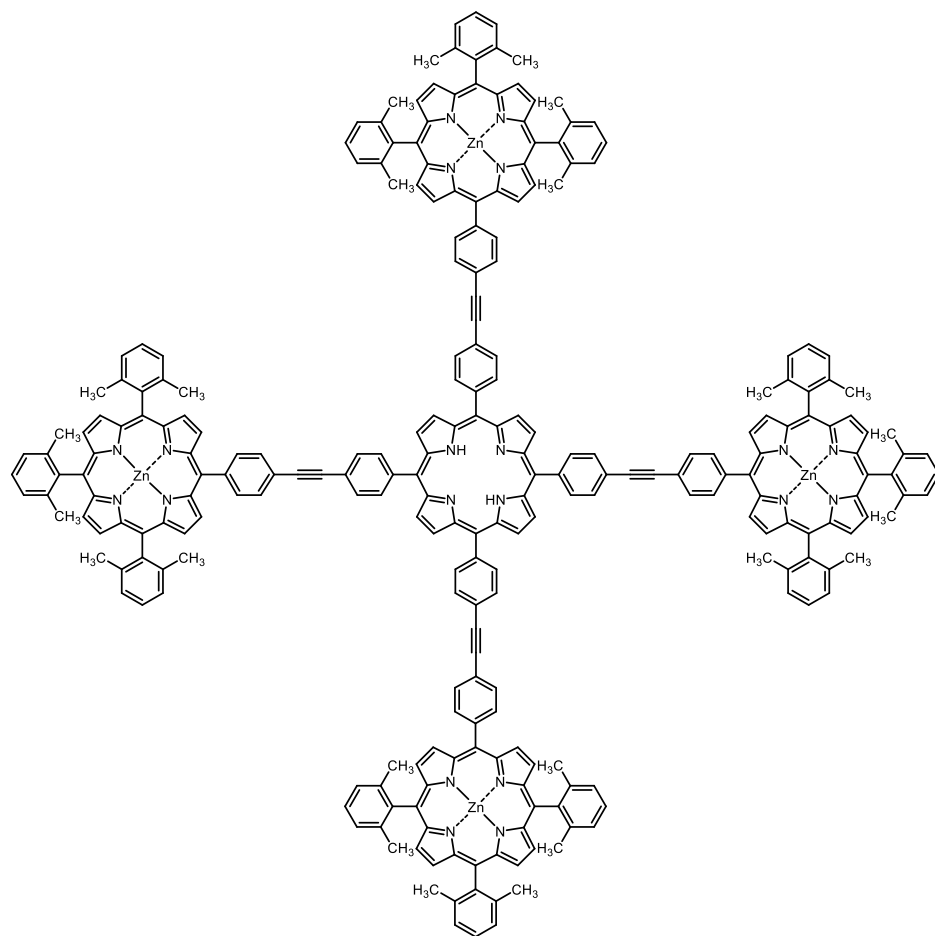
Porphyrin
(18π-System)
Substituenten R in
meso-Positionen

7. Porphyrin

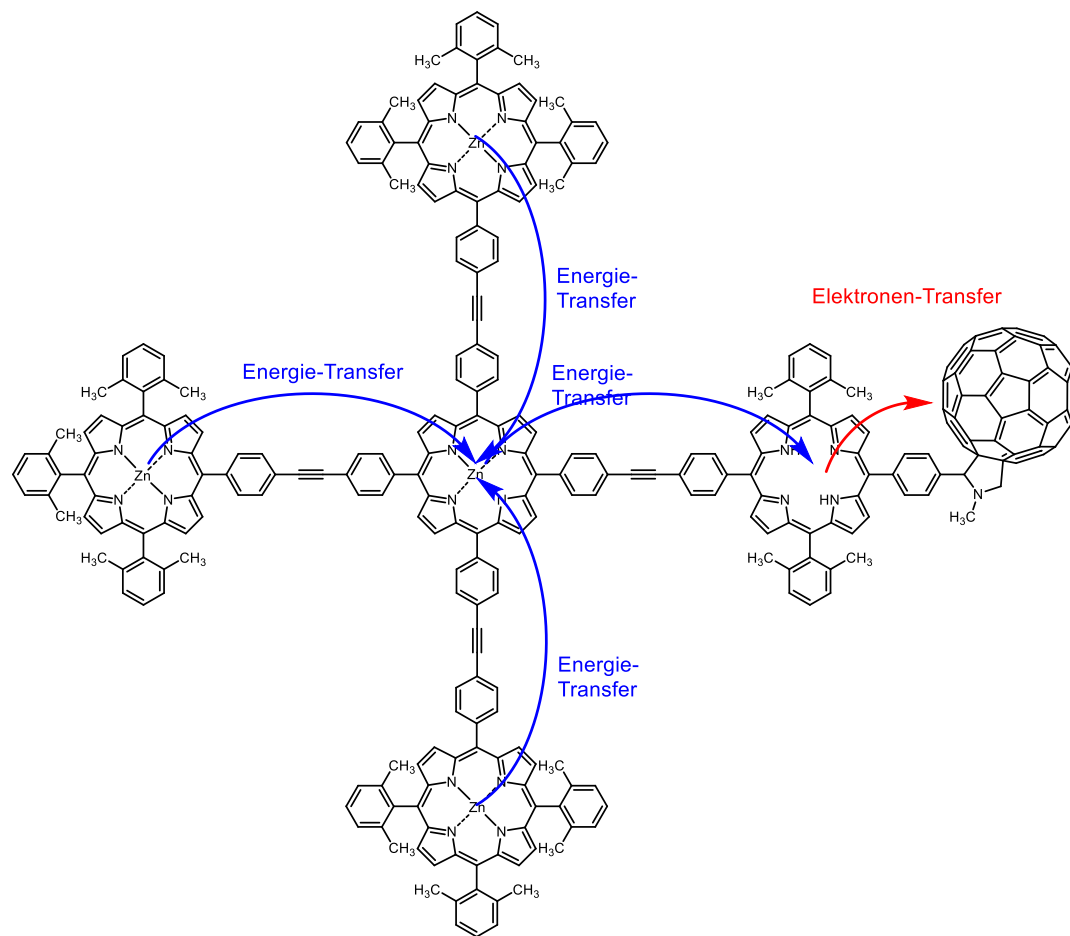


Chlorophyll a: $R = CH_3$
 Chlorophyll b: $R = CHO$

7. Porphyrin



Effizienter Energietransfer vom Zn-Porphyrin
zum inneren Porphyrin (95-99%)
J. Am. Chem. Soc. **1993**, 115, 7519.



Lebensdauer des ladungstrennten Zustands: 240 ns
J. Phys. Chem. A **2002**, 106, 2036.